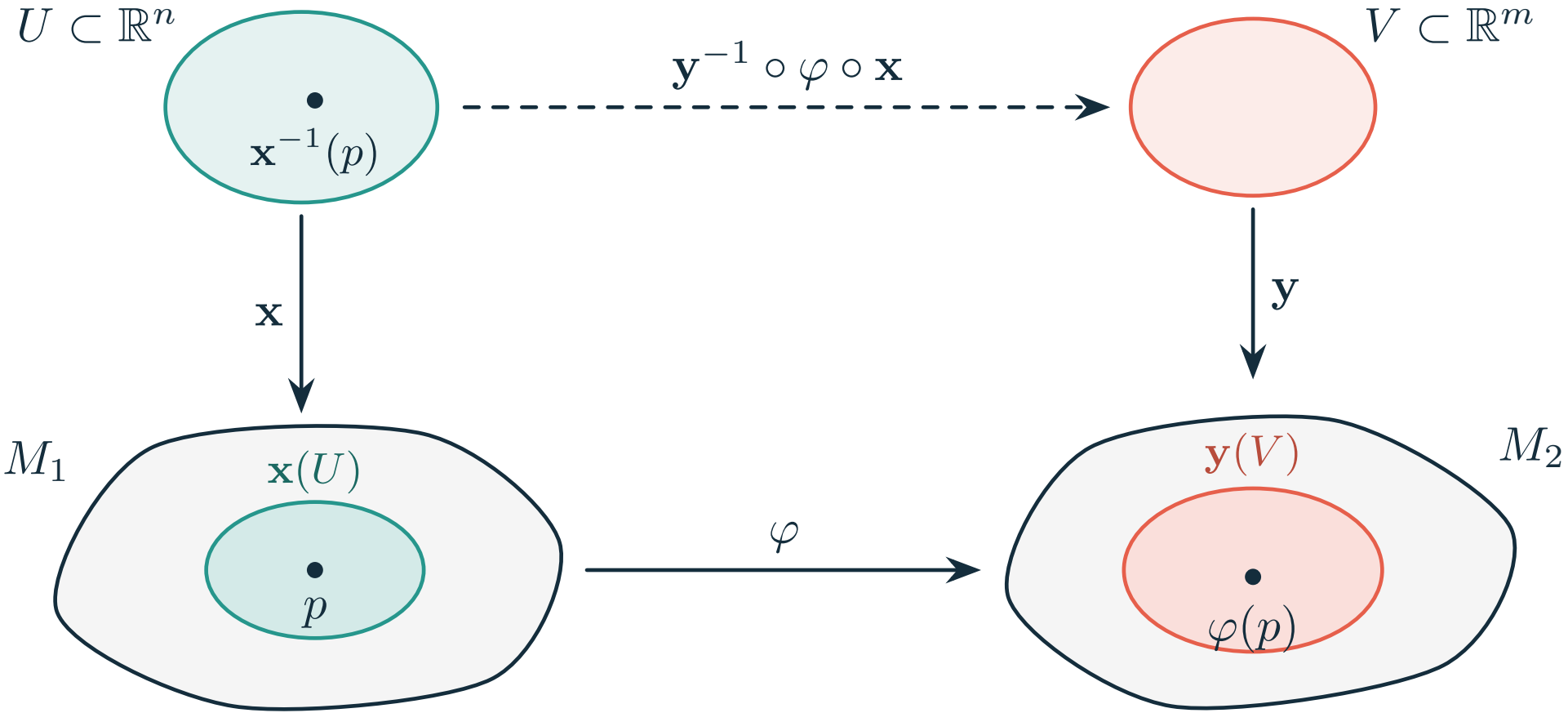

$U \subset \mathbb{R}^n$
$\mathbf{x}^{-1}(p)$
$\mathbf{y}^{-1} \circ \varphi \circ \mathbf{x}$
$V \subset \mathbb{R}^m$
$\mathbf{x}$
$\mathbf{y}$
M_1
$\mathbf{x}(U)$
p
φ
$\mathbf{y}(V)$
M_2
$\varphi(p)$